

Dictionary (dict) : Its the collection of keys and values pairs and seperation by :and enclosed with curly brackets {}

Dict is mutable datatype.

Value can be duplicate but key cant be duplicate.

There is no index concept in dict.

```
In [4]: d = {1:'One',2:'Two',3:'Three'} # dict with integers
```

```
In [5]: d
```

```
Out[5]: {1: 'One', 2: 'Two', 3: 'Three'}
```

```
In [7]: d = dict({1:'One',2:'Two',3:'Three'}) # created with using dict()
```

```
In [5]: d
```

```
Out[5]: {1: 'One', 2: 'Two', 3: 'Three'}
```

```
In [9]: d.items()
```

```
Out[9]: dict_items([(1, 'One'), (2, 'Two'), (3, 'Three')])
```

```
In [6]: d[4] = 'four' # Adding item to dict.  
d
```

```
Out[6]: {1: 'One', 2: 'Two', 3: 'Three', 4: 'four'}
```

```
In [8]: d[2+3j] = True # It adds the item to dict  
d
```

```
Out[8]: {1: 'One', 2: 'Two', 3: 'Three', 4: 'four', (2+3j): True}
```

```
In [ ]: # Remove:
```

```
In [9]: mydict = {1:'one' , 2:'two' , 3:'three' , 4:'four'}  
mydict
```

```
Out[9]: {1: 'one', 2: 'two', 3: 'three', 4: 'four'}
```

```
In [11]: mydict.pop(1) # Here 1: one item is removed from the dict
```

```
Out[11]: 'one'
```

```
In [12]: mydict
```

```
Out[12]: {2: 'two', 3: 'three', 4: 'four'}
```

popitem() : It removes random item for the dict.

```
In [13]: mydict.popitem()
```

```
Out[13]: (4, 'four')
```

```
In [26]: mydict1 = {1:'one' , 2:'two' , 3:'three' , 4:'four'}  
mydict1
```

```
Out[26]: {1: 'one', 2: 'two', 3: 'three', 4: 'four'}
```

```
In [28]: mydict = mydict1 # It creates the reference for mydict
```

```
In [29]: id(mydict), id(mydict1) # it has the same memoery for both the id
```

```
Out[29]: (2976937762624, 2976937762624)
```

```
In [31]: mydict1 = mydict.copy() # It creates the copy of dictionary
```

```
In [ ]:
```